Assignment 2: Naive Bayes Classification Report

1 Summary

This report documents the implementation and experimental study of two Naive Bayes classifiers on the Cleveland subset of the UCI Heart Disease dataset. The notebook implements Multinomial Naive Bayes and Multivariate Bernoulli Naive Bayes from scratch, converts continuous clinical measurements into discrete events using two training-derived binning strategies, handles missing values explicitly, and evaluates the models with five-fold stratified cross-validation.

The main observations are that (i) equal-width binning gives the strongest five-class results among the four principal model/binning combinations; (ii) Multinomial Naive Bayes with equal-width bins achieves the highest five-class macro-F1 (0.3546) and accuracy (0.5842); (iii) merging disease severity classes 1–4 into a single disease-present class substantially improves performance, with Bernoulli/equal-width reaching an F1 of about 0.837; (iv) empirical priors outperform uniform priors; (v) smoothing strength affects the two models differently; and (vi) Multinomial produces lower multiclass log loss than Bernoulli for both binning strategies.

2 Dataset and Preprocessing

The project loads the Cleveland data from the UCI Heart Disease dataset. The dataset contains 303 patient records and 14 attributes, including the target variable `num`. The target is retained as five classes: 0 denotes no presence of heart disease, while 1–4 represent increasing disease severity. The target attribute is not included in the input token representation.

Item	Implementation / Recorded Value
Dataset	UCI Heart Disease — Cleveland subset
Records	303
Attributes including target	14
Input features	13
Target	`num`
Target classes	0, 1, 2, 3, 4
Random seed	42
Cross-validation	5-fold stratified

2.1 Class Distribution

Class	Meaning	Count
0	No presence of heart disease	164
1	Presence, severity 1	55
2	Presence, severity 2	36
3	Presence, severity 3	35
4	Presence, severity 4	13
Total		303

The target is imbalanced, with class 0 containing 164 records and class 4 containing only 13. This imbalance is particularly relevant for macro-F1 and for the prior-probability experiments.

2.2 Missing Values

The source data uses "?" for missing values. The notebook converts these entries to an explicit missing representation rather than silently dropping them. Missingness occurs only in `ca` and `thal`.

Feature	Missing records
ca	4
thal	2
All other input features	0

The main implementation retains these observations by generating explicit tokens such as ca=missing or thal=missing. A second experiment removes incomplete records and compares macro-F1 with the explicit-token strategy.

3 Feature Representation and Preprocessing

Naive Bayes is required to operate on discrete events. The notebook therefore keeps categorical attributes as categorical feature-value events and discretises the five continuous attributes age, trestbps, chol, thalach, and oldpeak.

Feature type	Attributes / treatment
Categorical	sex, cp, fbs, restecg, exang, slope, ca, thal
Continuous → bins	age, trestbps, chol, thalach, oldpeak
Number of bins	10
Missing representation	Explicit feature=missing token
Patient representation	Bag/list of feature-value tokens

3.1 Equal-width and Quantile Binning

Two alternative binning strategies are implemented. Equal-width binning divides the training range of a continuous feature into ten equal-width intervals. Quantile binning uses training quantiles to form ten approximately equally populated intervals, with duplicate boundaries removed where necessary.

Crucially, the thresholds are learned only from the training portion of each fold and are then applied unchanged to the corresponding held-out records.

A patient is consequently represented by tokens such as age_bin=2 or cp=4. For this structured dataset, the tokenization/vocabulary requirement is realised through explicit feature-value token construction rather than natural-language text processing.

4 Methodology

Both classifiers are implemented manually without calling a pre-built Naive Bayes classifier. The implementation uses class counts, vocabulary construction, Laplace smoothing, logarithmic probabilities, and deterministic prediction from the resulting class scores.

4.1 Multinomial Naive Bayes

The Multinomial model counts feature-value tokens within each class. With vocabulary size V, token count $N(t, c)$, total token count $N(c)$, and smoothing parameter α, the smoothed likelihood is

$$P(t \mid c) = \frac{N(t, c) + \alpha}{N(c) + \alpha V}.$$

For a patient represented by tokens T, the prediction score is computed in log space as the log prior plus the sum of log likelihoods for the patient's tokens. This avoids numerical underflow from multiplying many small probabilities.

4.2 Multivariate Bernoulli Naive Bayes

The Bernoulli model represents each vocabulary token as a binary event: present or absent. The implementation therefore evaluates both events across the full vocabulary rather than only the tokens that occur in a patient record.

For a present token,

$$P(t \text{ present} \mid c) = \frac{N_{\text{present}}(t, c) + \alpha}{N(c) + 2\alpha}.$$

For an absent token,

$$P(t \text{ absent} \mid c) = \frac{N(c) - N_{\text{present}}(t, c) + \alpha}{N(c) + 2\alpha}.$$

This explicit treatment of absent vocabulary events distinguishes the multivariate Bernoulli model from a token-count-only implementation.

4.3 Priors, Smoothing, and Deterministic Prediction

The default experiments use empirical class priors and $\alpha = 1.0$ (Laplace/add-one smoothing). The notebook additionally evaluates uniform priors and α values 0.5 and 2.0.

Class scores are compared in log space, and the class array is deterministic, so the argmax operation resolves equal-score cases deterministically.

Configuration	Values tested
Class priors	Empirical, uniform
Smoothing α	0.5, 1.0, 2.0
Binning	Equal-width, quantile
Target formulation	Five-class; binary 0 vs 1–4

5 Training and Evaluation Protocol

The notebook creates five stratified folds using a fixed random seed of 42. The fold sizes are 62, 61, 61, 60, and 59 records. Stratification keeps the class composition approximately consistent across folds.

Fold	Size	Class distribution
1	62	0:33, 1:11, 2:8, 3:7, 4:3
2	61	0:33, 1:11, 2:7, 3:7, 4:3
3	61	0:33, 1:11, 2:7, 3:7, 4:3
4	60	0:33, 1:11, 2:7, 3:7, 4:2
5	59	0:32, 1:11, 2:7, 3:7, 4:2

For every fold, binning thresholds and model probabilities are fitted on the training portion and then evaluated on the held-out fold. The reported metrics include accuracy, precision, recall, F1, macro-F1, micro-F1, confusion matrices, precision-recall curves, and ROC curves.

6 Task 1: Main Five-Class Experimental Results

Model	Binning	Accuracy	Precision	Recall	Macro F1	Micro F1
Multinomial	Equal-width	0.5842	0.3934	0.3492	0.3546	0.5842
Bernoulli	Equal-width	0.5809	0.3872	0.3463	0.3500	0.5809
Multinomial	Quantile	0.5545	0.3505	0.3094	0.3137	0.5545
Bernoulli	Quantile	0.5611	0.3929	0.3208	0.3255	0.5611

The equal-width strategy is stronger than quantile binning for both classifiers in the main five-class experiment. Multinomial/equal-width gives the highest accuracy and macro-F1, while Bernoulli/equal-width is a close second.

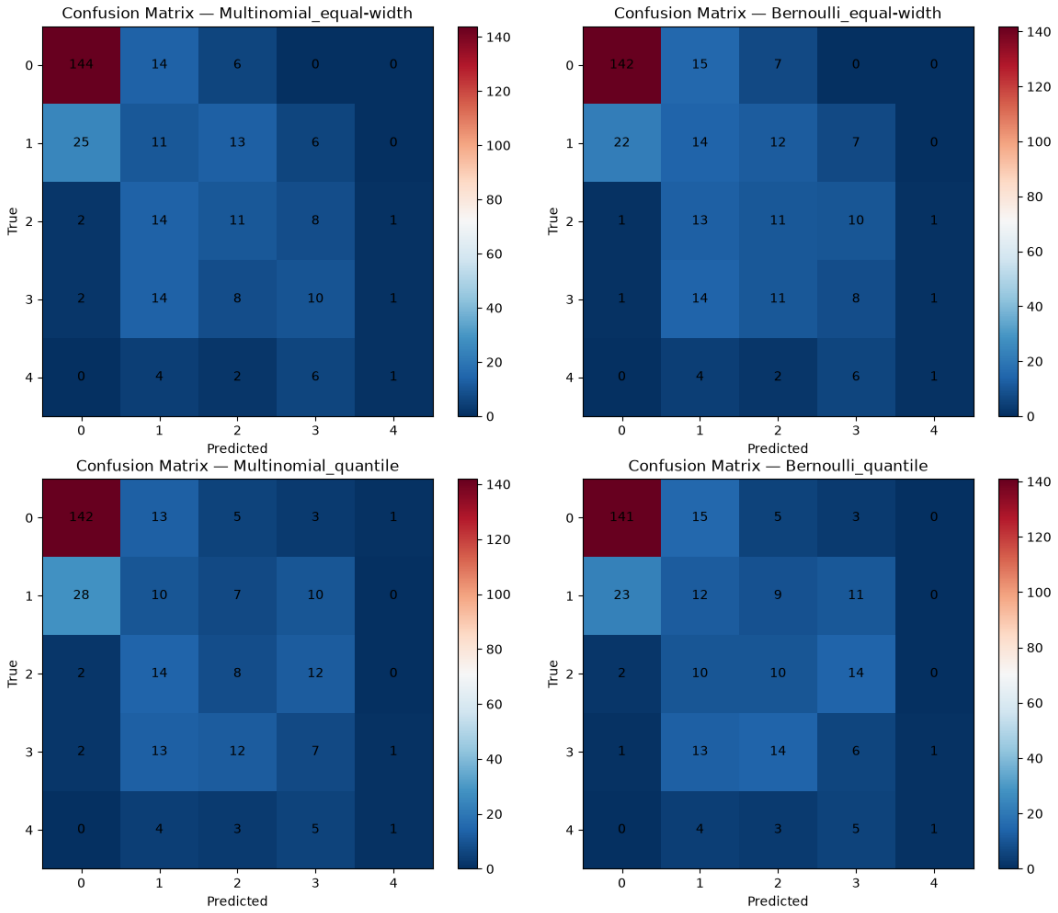


Figure 1: Five-class confusion matrices for the four principal model/binning configurations.

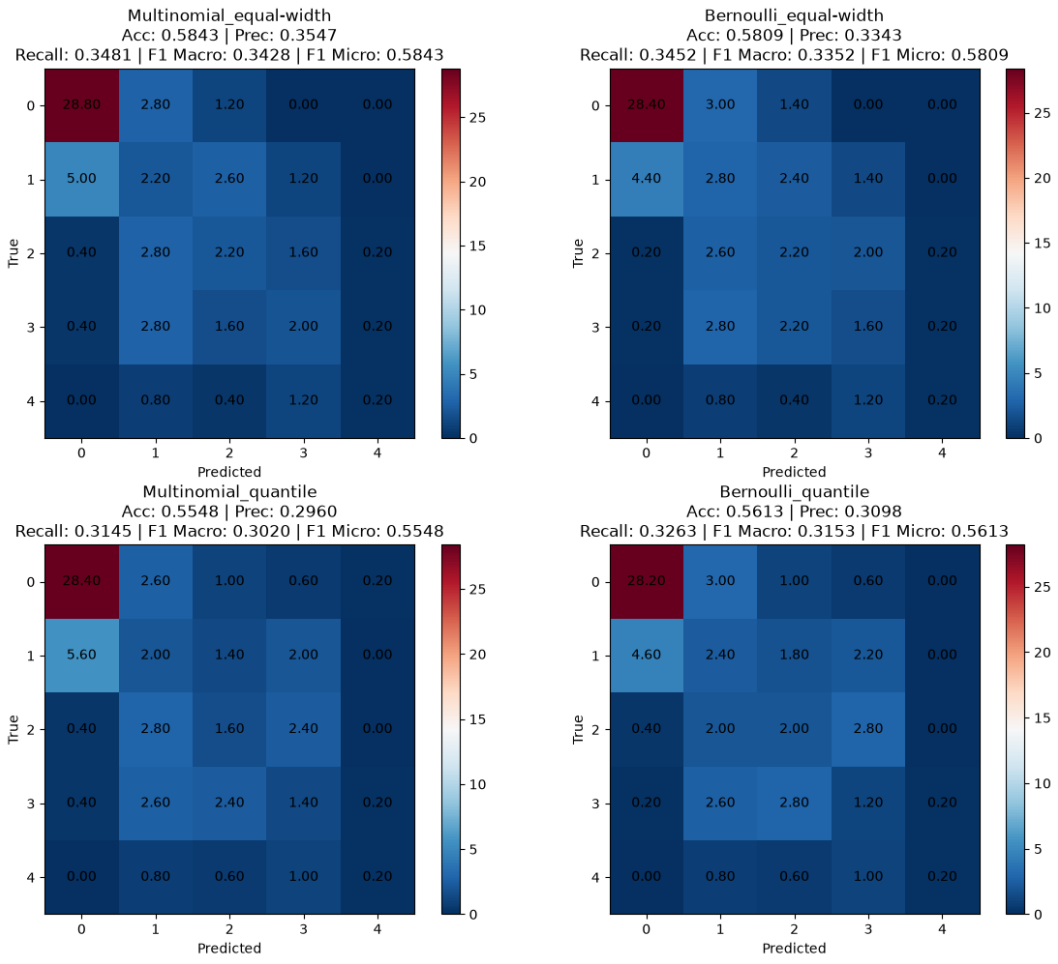


Figure 2: Five-class validation metrics across the principal configurations.

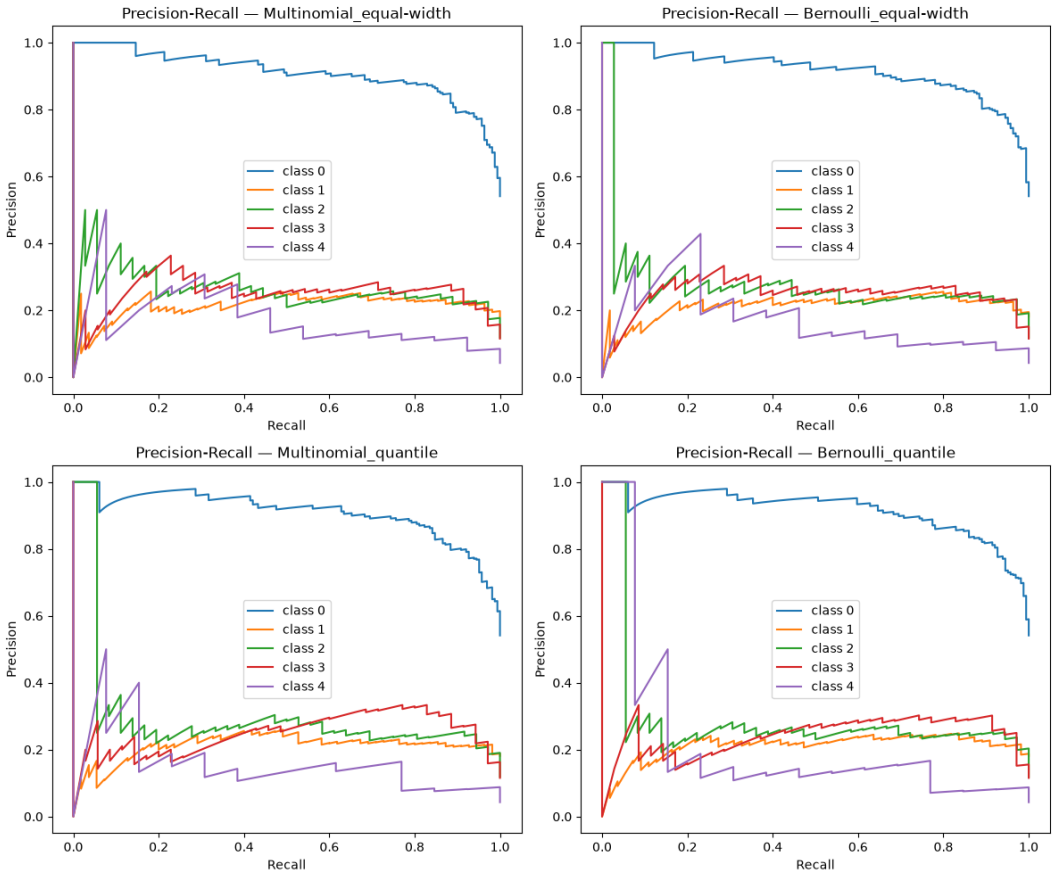


Figure 3: Precision–Recall curves for the four principal configurations.

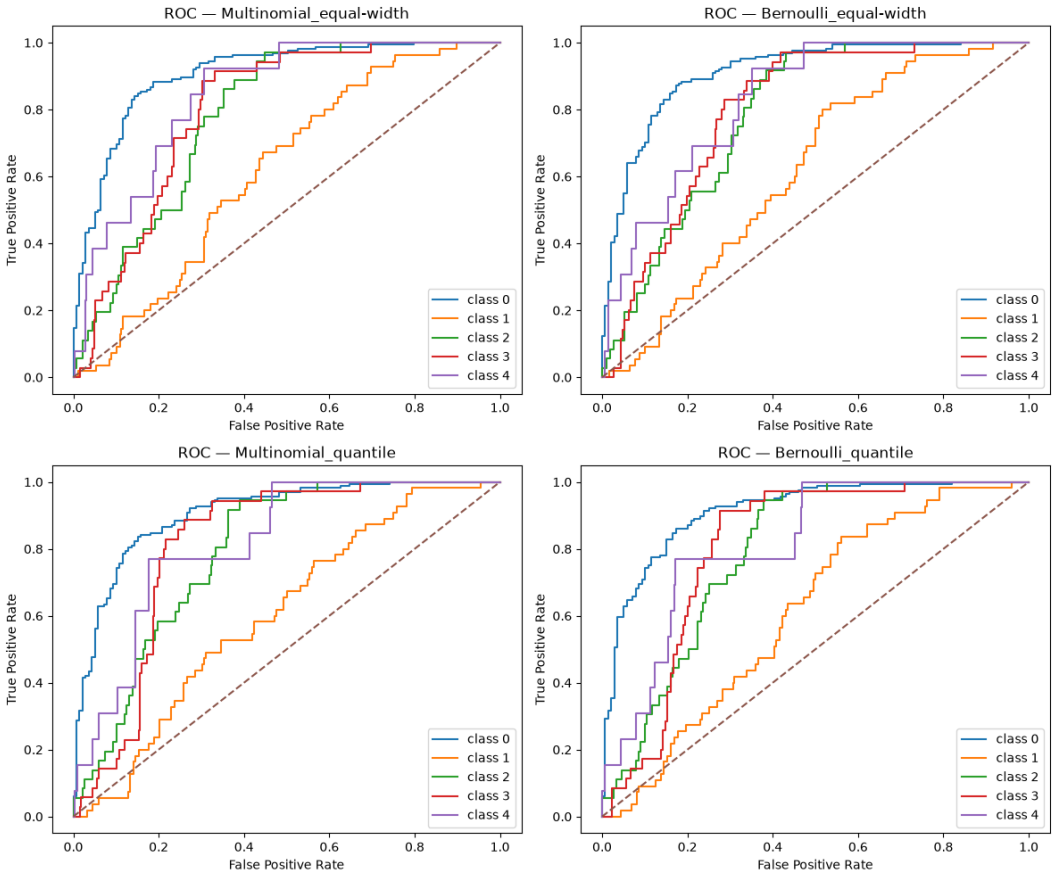


Figure 4: ROC curves for the four principal configurations.

7 Task 2: Five-Class versus Binary Disease Classification

To examine the effect of class imbalance and severity granularity, classes 1–4 are merged into a single disease-present class while class 0 remains disease absent.

Model	Binning	Accuracy	Precision	Recall	F1
Multinomial	Equal-width	0.8352	0.8361	0.8331	0.8335
Bernoulli	Equal-width	0.8386	0.8389	0.8379	0.8375
Multinomial	Quantile	0.8321	0.8332	0.8296	0.8301
Bernoulli	Quantile	0.8256	0.8252	0.8236	0.8241

The binary task is much easier than the five-class severity problem. The best observed result is Bernoulli with equal-width bins ($F1 = 0.8375$). The improvement shows that much of the difficulty in the original problem comes from distinguishing among four disease-severity categories that are individually much smaller than class 0.

8 Effect of Binning Strategy

Switching between equal-width and quantile binning changes the discrete feature tokens and therefore changes the class-conditional likelihoods. The notebook records 57 changed predictions for Multinomial Naive Bayes and 46 for Bernoulli Naive Bayes.

A concrete example is patient index 1. Under equal-width binning, both models predict class 1, whereas under quantile binning both predict class 3. The actual target is class 2.

Feature	Patient value
age	67
sex	1
cp	4
trestbps	160
chol	286
fbs	0
restecg	2
thalach	108
exang	1
oldpeak	1.5
slope	2
ca	3
thal	3
True class	2

This example demonstrates that binning is not a neutral preprocessing step: the same continuous measurement may map to different events under the two strategies, leading to different posterior scores and predictions.

9 Missing-Value Experiment

Model	Explicit missing token: Macro F1	Remove incomplete records: Macro F1	Records retained
Multinomial	0.3137	0.3400	297 / 303
Bernoulli	0.3255	0.3588	297 / 303

Removing the six incomplete observations produces higher macro-F1 for both models in the recorded experiment. The increase is especially visible for Bernoulli Naive Bayes. However, the result should not be interpreted as a

general rule that deletion is always preferable: only six records are affected, and an explicit missing token preserves information that would otherwise be discarded.

10 Posterior Score Analysis

The notebook examines two validation patients using Multinomial Naive Bayes with quantile binning. One patient is correctly classified as class 0, while another true class-0 patient is misclassified as class 1.

Class	Correct: log score	Correct: posterior	Misclassified: log score	Misclassified: posterior
0	-49.9389	0.9025	-53.4615	0.1396
1	-52.2763	0.0872	-51.7725	0.7556
2	-54.8964	0.0063	-54.1041	0.0734
3	-55.3925	0.0039	-55.5466	0.0173
4	-58.4598	0.0002	-55.7517	0.0141

For the correctly classified patient, the posterior probability of class 0 is approximately 0.9025, far above the other classes. For the misclassified patient, class 1 receives approximately 0.7556 posterior probability even though the true class is 0.

11 Task 3: Empirical versus Uniform Priors

Model	Prior	Macro F1	Accuracy
Multinomial	Empirical	0.3137	0.5545
Multinomial	Uniform	0.3037	0.5215
Bernoulli	Empirical	0.3255	0.5611
Bernoulli	Uniform	0.3074	0.5149

Empirical priors perform better for both models. Uniform priors remove the information contained in the observed class frequencies and consequently reduce both accuracy and macro-F1.

12 Smoothing Experiment

Model	$\alpha = 0.5$	$\alpha = 1.0$	$\alpha = 2.0$
Multinomial Macro F1	0.3054	0.3137	0.2900
Bernoulli Macro F1	0.3389	0.3255	0.3009

The best tested α differs between the two models. Multinomial reaches its highest recorded macro-F1 at $\alpha = 1.0$, whereas Bernoulli performs best at $\alpha = 0.5$. Increasing α to 2.0 reduces macro-F1 for both models.

13 Posterior Probabilities and Calibration

Model	Binning	Multiclass log loss
Multinomial	Equal-width	1.1472
Bernoulli	Equal-width	1.4388
Multinomial	Quantile	1.1587
Bernoulli	Quantile	1.4368

The notebook converts log scores into normalised posterior probabilities using a numerically stable score-shifting procedure before exponentiation. Multinomial Naive Bayes has lower multiclass log loss than Bernoulli for both binning strategies.

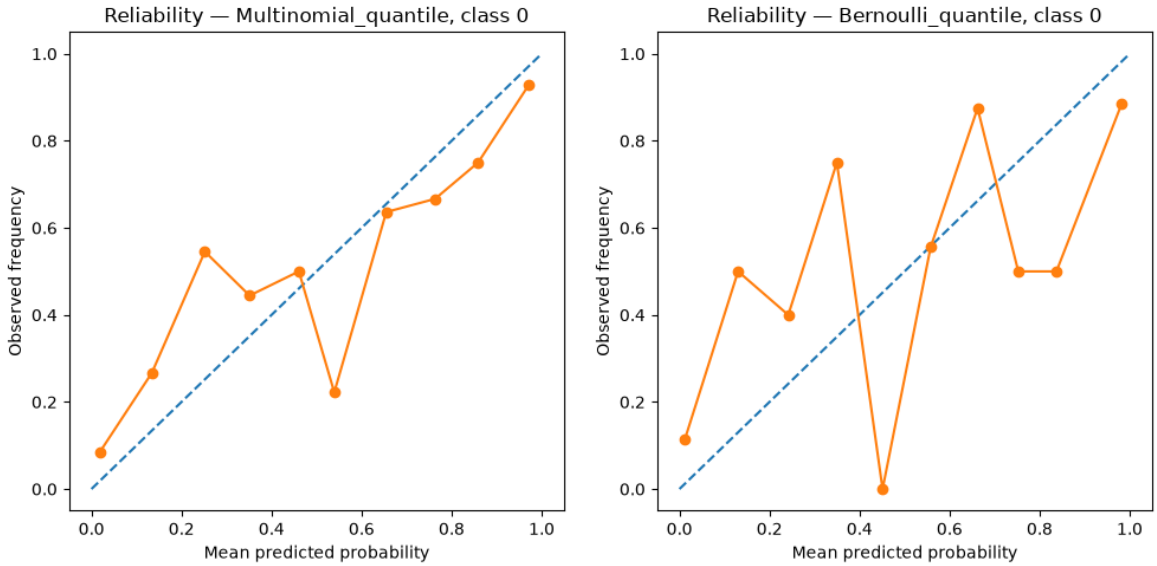


Figure 5: One-vs-rest reliability diagrams for class 0.

14 Sensitivity to an Irrelevant Shuffled Feature

Model	Baseline Macro F1	With shuffled token	Change
Multinomial	0.3137	0.3283	+0.0146
Bernoulli	0.3255	0.3398	+0.0143

A shuffled feature-value token column with five possible values is generated using the fixed random seed. The token has no clinical meaning, yet the measured macro-F1 changes slightly for both models. Because the dataset is small, finite-sample associations can arise by chance, allowing an irrelevant feature to alter likelihood estimates and predictions.

15 Overall Analysis and Discussion

- Equal-width binning provides the strongest five-class results.
- Multinomial/equal-width gives the highest five-class accuracy (0.5842) and macro-F1 (0.3546).
- Binary disease-present classification is substantially easier than retaining all five severity classes.
- Empirical class priors outperform uniform priors.
- The preferred smoothing strength is model-dependent: $\alpha = 1.0$ for Multinomial and $\alpha = 0.5$ for Bernoulli.
- Multinomial produces lower multiclass log loss than Bernoulli.
- Explicitly modelling missing values preserves all 303 observations.
- The irrelevant-token experiment shows that finite-sample likelihood estimation can be affected by an irrelevant feature.

16 Reproducibility and Implementation Checklist

Item	Recorded configuration
Random seed	42
Number of folds	5
Binning	10 bins
Binning strategies	Equal-width, quantile
Default smoothing	$\alpha = 1.0$
Additional smoothing	$\alpha = 0.5, 2.0$
Priors	Empirical, uniform
Target experiments	Five-class; binary 0 vs 1–4
Missing strategies	Explicit token; record removal comparison
Models	Manual MultinomialNB; manual BernoulliNB

17 Limitations and Notes

- The Cleveland dataset is small (303 records), so macro-F1 and calibration estimates can vary across folds.
- The five-class target is highly imbalanced, particularly for class 4.
- The irrelevant-feature experiment is a single fixed-seed shuffle rather than a multi-run significance study.
- The structured dataset does not naturally require natural-language tokenisation or stopword removal; feature-value token construction is used to satisfy the discrete-event formulation.

18 Conclusion

The experiments show that both manually implemented Naive Bayes models can successfully perform heart-disease classification, but performance depends strongly on the target formulation, binning strategy, class priors, and handling of missing values. The key results required by the assignment are summarized below.

Table 1: Summary of Important Experimental Results

Experiment	Multinomial NB	Bernoulli NB
5-class, Equal-width Accuracy	0.5842	0.5809
5-class, Equal-width Macro-F1	0.3546	0.3500
5-class, Equal-width Micro-F1	0.5842	0.5809
5-class, Quantile Accuracy	0.5545	0.5611
5-class, Quantile Macro-F1	0.3137	0.3255
Binary, Equal-width Accuracy	0.8352	0.8386
Binary, Equal-width F1	0.8335	0.8375
Binary, Quantile Accuracy	0.8321	0.8256
Binary, Quantile F1	0.8301	0.8241

Merging classes 1–4 into a single disease-present class substantially improved performance compared with the original five-class problem, mainly because the five-class setting is affected by severe class imbalance. Equal-width binning generally performed better than quantile binning for the five-class Multinomial model, while the binning choice changed predictions for 57 patients for Multinomial NB and 46 patients for Bernoulli NB; for example, patient 1 changed from class 1 to class 3 while the true class was 2.

Explicit missing-value tokens produced Macro-F1 scores of 0.3137 (Multinomial) and 0.3255 (Bernoulli), compared with 0.3400 and 0.3588 after removing incomplete records, respectively. However, only 297 of the 303 records were complete, so retaining an explicit missing token preserves the available data and follows the assignment requirement.

Empirical priors outperformed uniform priors: Multinomial Macro-F1 decreased from 0.3137 to 0.3037 and accuracy from 0.5545 to 0.5215 with uniform priors, while Bernoulli Macro-F1 decreased from 0.3255 to 0.3074

and accuracy from 0.5611 to 0.5149. Thus, accuracy alone can hide the effect of class imbalance, making Macro-F1 important for evaluating the minority classes.

Posterior analysis also demonstrated that the models produce meaningful class confidence. For a correctly classified patient (index 7), the predicted class 0 had normalized posterior probability 0.9025, whereas for a misclassified patient (index 20), class 1 received the highest probability of 0.7556 despite the true class being 0.

Using numerically stable normalized posteriors, the multiclass log losses were 1.1472 for Multinomial equal-width, 1.1587 for Multinomial quantile, 1.4388 for Bernoulli equal-width, and 1.4368 for Bernoulli quantile. Finally, adding a shuffled irrelevant feature did not substantially degrade performance; Macro-F1 changed by only +0.0146 for Multinomial NB and +0.0143 for Bernoulli NB, indicating limited sensitivity to this irrelevant feature under the fixed random seed. Overall, the experiments demonstrate the importance of class balance, feature discretization, missing-value treatment, prior selection, smoothing, and calibrated posterior probabilities when applying Naive Bayes to clinical classification.

19 Team Contributions

Member	**Contribution**
Anshul Kaushal (2611AI10)	Implementation and experimental development.
Aditya Sharma (2611CS11)	Implementation and experimental development.
Roland Zohmingthana (2611MC03)	Preparation and development of the report.